

Emergency Care AI Coordination PoC - Checklist

Project Orientation

1. Clarify scope: Assistive, non-diagnostic, PoC/demo only.
2. Stay close to APONA scaffold for comparability.
3. Focus on human well-being and workload reduction, not just logistics.

Workflow Stages

1. Pre-arrival intake from IVENA (EMS triage, pre-clinical data).
2. Arrival admin: scan insurance card, open admission file in ORBIS mock.
3. Order assist (staff approval required) for labs/imaging.
4. Capacity and bed assignment proposals from IVENA/KIS feed.
5. Handoff package creation for receiving department.

Model Design

1. GRU + MLP baseline (Mamba optional for long sequences).
2. Inputs: EMS triage, vitals, complaint, allergies, meds, capacity snapshot.
3. Outputs: Next best operational action + ETAs for key milestones.
4. Synthetic data with realistic timelines and distributions.

Safety & Guardrails

1. All proposals require clinician approval before execution.
2. Audit logging for all actions and approvals.
3. Staleness checks for capacity data (>N min triggers warning).
4. Conflict handling (IVENA vs KIS mismatch).
5. Safe defaults for missing/uncertain data.

Integration & Simulation

1. Replay IVENA feed or synthetic equivalent.
2. Mock ORBIS HL7 v2 interface (ADT, ORM, ORU).
3. Use Mirth Connect or local file queue for HL7 message flow.
4. Simulate realistic delays for lab/imaging results.

Evaluation Metrics

1. Admin minutes saved per patient.

2. Median and p90 time-to-first-result.
3. NASA-TLX workload survey results.
4. Percentage of overridden proposals.
5. System suggestion latency (<200ms).

Development Milestones

1. Week 1: Data generator + IVENA replay + HL7 bus + baseline rule-engine.
2. Week 2: Train GRU policy, wire approval flow, implement guardrails.
3. Week 3: UI polish, run 4-hour 50 patients/12h scenario, gather feedback.